

TECHSHIP INTERNSHIP

Day 8 – MongoDB Fundamentals & Database Operations

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Date : 10-06-2026

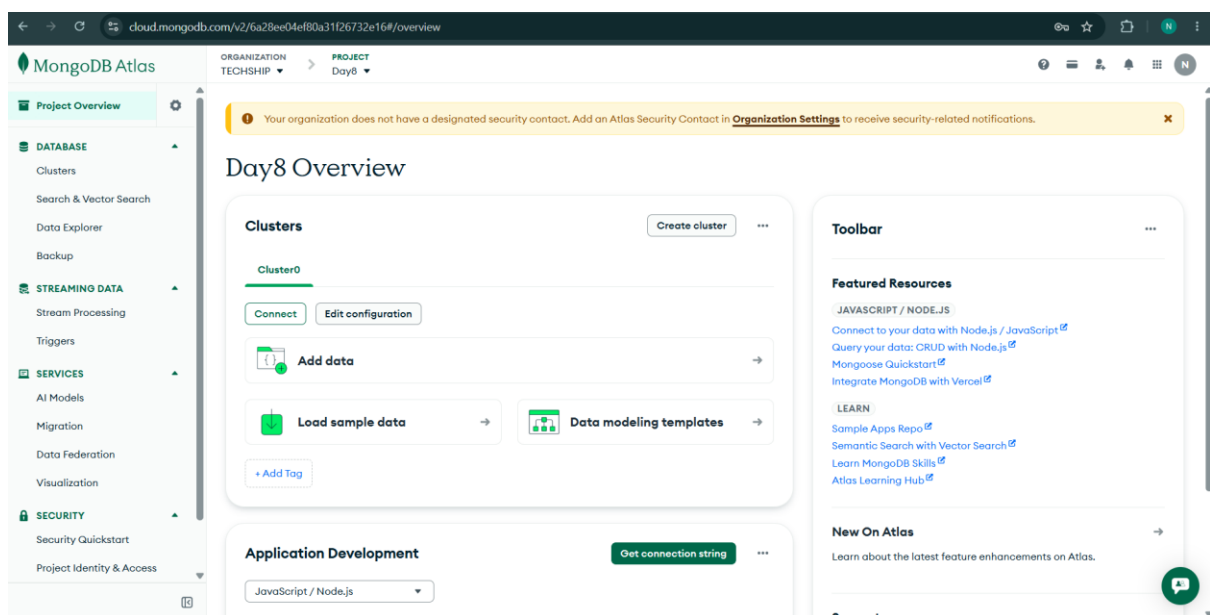
During Day 8 of the TECHSHIP Internship Program, I learned about databases and how they are used to store and manage data. I also learned the difference between SQL and MongoDB. MongoDB stores data in documents and collections, making it flexible and easy to use.

I used MongoDB Atlas to create databases and collections. I inserted records and performed CRUD operations such as Create, Read, Update, and Delete. These activities helped me understand how data is managed in applications.

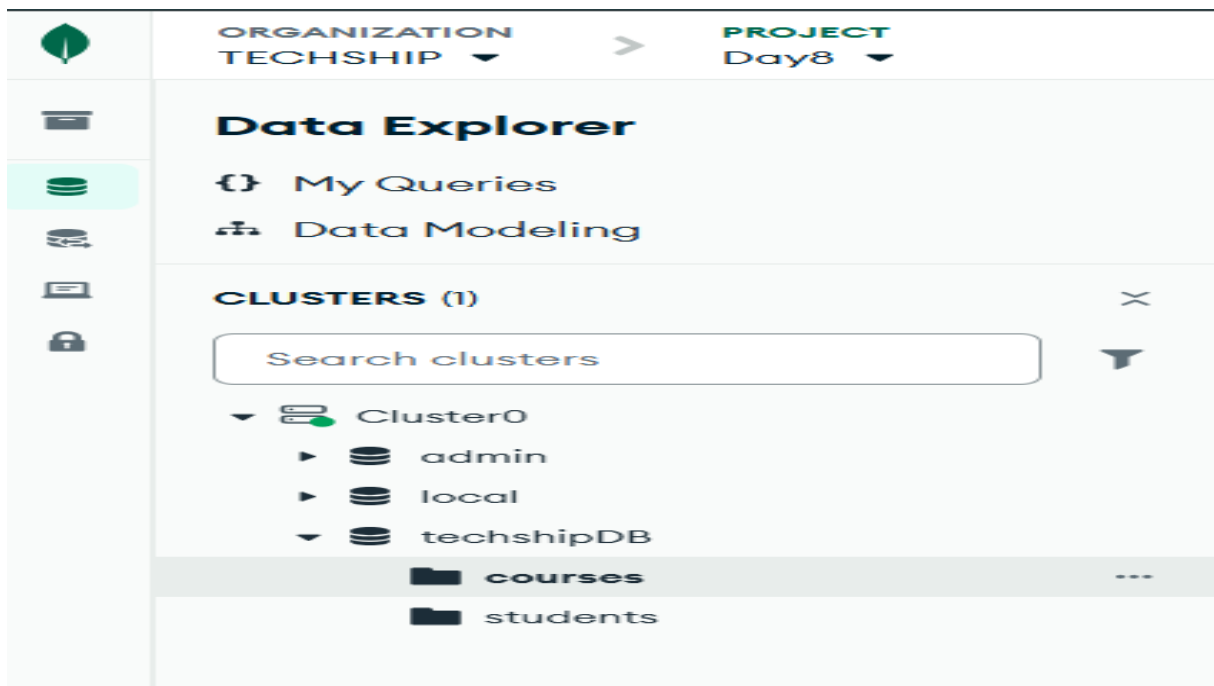
One challenge I faced was connecting MongoDB Atlas to VS Code. I had some difficulties at first, but I was able to solve them and complete the tasks successfully. Overall, this session improved my understanding of MongoDB and database management.

Activities

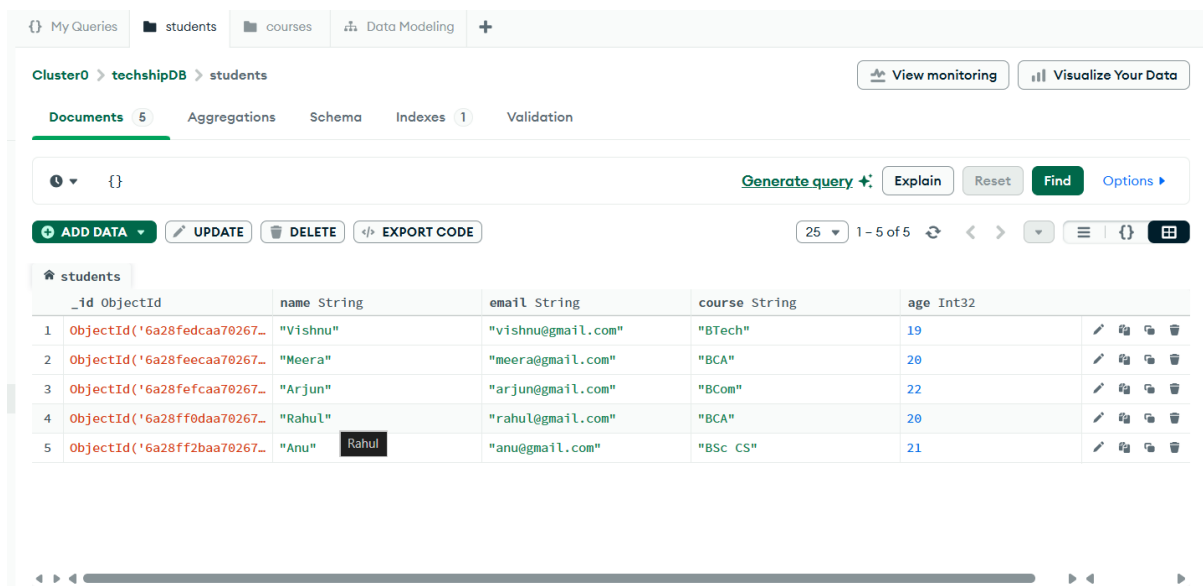
Activity 1 – Create a MongoDB Atlas Account



Activity 2 – Create a Database



Activity 3 – Insert Student Records



Activity 6 – Delete Records

Cluster0 > techshipDB > students

Documents 4 Aggregations Schema Indexes 1 Validation

Generate query [View monitoring](#) [Visualize Your Data](#)

[Explain](#) [Reset](#) [Find](#) [Options](#)

ADD DATA UPDATE DELETE EXPORT CODE

25 0 - 0 of 0

No results

Try modifying your query to get results.

Activity 7 – Design a Collection

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Cluster0

- admin
- local
- techshipDB
 - courses
 - students
 - tasks

Cluster0 > techshipDB > tasks

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[Explain](#) [Reset](#) [Find](#) [Options](#)

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	title String	description String	status String	
1	ObjectId('6a2902a7aa70267...')	"Complete Assignment"	"Finish MongoDB Activitie...	"Pending"	Edit Delete
2	ObjectId('6a2902b5aa70267...')	"Prepare Report"	"Write internship report"	"Pending"	Edit Delete
3	ObjectId('6a2902d8aa70267...')	"Review Notes"	"Study MongoDB concepts"	"Completed"	Edit Delete
4	ObjectId('6a2902efaa70267...')	"Update Records"	"Modify student data"	"Pending"	Edit Delete
5	ObjectId('6a2902fcaa70267...')	"Submit Work"	"Upload screenshots and r...	"Pending"	Edit Delete

Challenge Activities

Challenge 1 – Course Management Collection

My Queriesstudentstasks**courses**Data Modeling+

Cluster0 > techshipDB > courses

View monitoringVisualize Your Data

Documents 5AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATAUPDATEDELETEDEXPORT CODE

251 - 5 of 5

courses

	_id ObjectId	courseName String	duration String	instructor String	fees Int32	
1	ObjectId('6a2903a4aa70267...	"Web Development"	"6 Months"	"Arun Kumar"	15000	
2	ObjectId('6a2903b1aa70267...	"Python Programming"	"4 Months"	"Anjali Nair"	12000	
3	ObjectId('6a2903c0aa70267...	"Data Science"	"8 Months"	"Rahul Menon"	25000	
4	ObjectId('6a2903cfaa70267...	"Java Programming"	"5 Months"	"Priya Das"	14000	
5	ObjectId('6a2903e6aa70267...	"Mobile App Development"	"6 Months"	"Vishnu Raj"	18000	

Challenge 2 – Employee Database

Create Operation

Data Explorer

My QueriesData Modeling

CLUSTERS ()

Search clusters

Cluster0

- admin
- local
- techshipDB
 - courses
 - employees**
 - students
 - tasks

My Queriesstudentstasks**courses****employees**Data Modeling+

Cluster0 > techshipDB > employees

View monitoringVisualize Your Data

Documents 5AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATAUPDATEDELETEDEXPORT CODE

251 - 5 of 5

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2904a0aa70267...	"Akhil"	"Designing"	35000	2	
2	ObjectId('6a2904c0aa70267...	"Meera"	"HR"	30000	3	
3	ObjectId('6a2904cfaa70267...	"Rahul"	"Finance"	40000	4	
4	ObjectId('6a2904ddaa70267...	"Anu"	"Marketing"	32000	2	
5	ObjectId('6a2904efaa70267...	"Vishnu"	"IT"	45000	5	

Read Operation

Cluster0 > techshipDB > employees View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query ↗ Explain Reset Find Options ▶

ADD DATA ▼ UPDATE DELETE EXPORT CODE 25 1 - 1 of 1 ◀ ▶ ⌵ ⌶ ⌵

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2904aeaa70267...	"Akhil"	"Designing"	35000	2	✎ 📄 🗑️

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Update Operation

Cluster0 > techshipDB > employees View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query ↗ Explain Reset Find Options ▶

ADD DATA ▼ UPDATE DELETE EXPORT CODE 25 1 - 1 of 1 ◀ ▶ ⌵ ⌶ ⌵

employees

	_id ObjectId	employeeName String	department String	salary Int32	experience Int32	
1	ObjectId('6a2904aeaa70267...	"Akhil"	"Designing"	50000	5	✎ 📄 🗑️

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Delete Operation

Cluster0 > techshipDB > employees View monitoring Visualize Your Data

Documents 4 Aggregations Schema Indexes 1 Validation

Generate query ↗ Explain Reset Find Options ▶

ADD DATA ▼ UPDATE DELETE EXPORT CODE 25 0 - 0 of 0 ◀ ▶ ⌵ ⌶ ⌵



No results

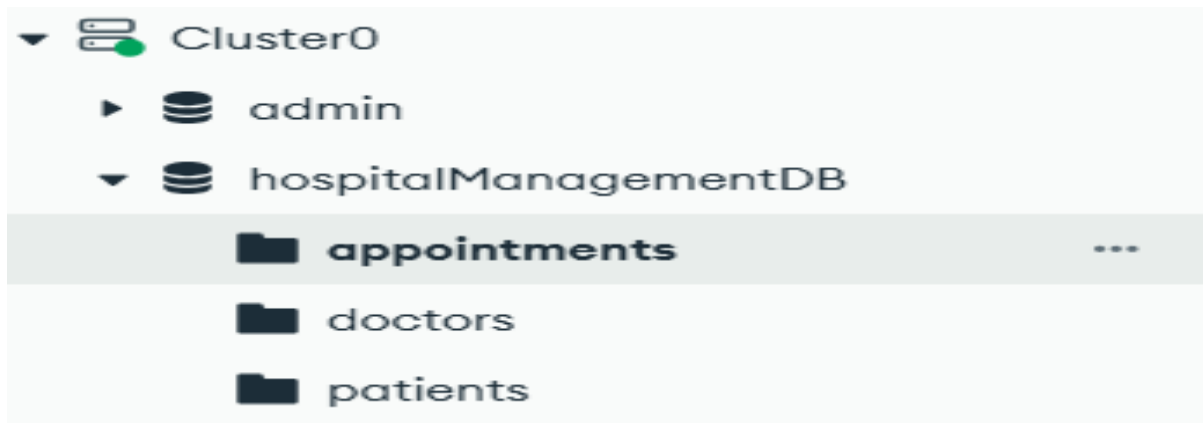
Try modifying your query to get results.

Challenge 3 – Design Your Own Database

Hospital Management System

Collections created:

1. Appointments
2. Doctors
3. Patients



Appointments

Cluster0 > hospitalManagementDB > appointments [View monitoring](#) [Visualize Your Data](#)

[Documents](#) 3 [Aggregations](#) [Schema](#) [Indexes](#) 1 [Validation](#)

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#) 25 1 - 3 of 3 [Refresh](#) [Filter](#) [Columns](#) [Grid](#)

	_id ObjectId	appointmentId String	patientName String	doctorName String	date String	
1	ObjectId('6a2907b5aa70267...	"A101"	"Rahul Kumar"	"Dr. Anjali"	"2026-06-10"	Edit Duplicate Delete
2	ObjectId('6a2907d2aa70267...	"A102"	"Meera Nair"	"Dr. Vishnu"	"2026-06-12"	Edit Duplicate Delete
3	ObjectId('6a2907dfaa70267...	"A103"	"Arjun Das"	"Dr. Priya"	"2026-06-15"	Edit Duplicate Delete

Navigation: < > < > < >

Doctors

Cluster0 > hospitalManagementDB > doctors

Documents 3

Aggregations

Schema

Indexes 1

Validation

View monitoring

Visualize Your Data

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 3 of 3

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doctors

	_id ObjectId	doctorId String	name String	specialization String	experience Int32	
1	ObjectId('6a290769aa70267...	"D101"	"Dr. Anjali"	"General Medicine"	8	
2	ObjectId('6a290778aa70267...	"D102"	"Dr. Vishnu"	"Neurology"	10	
3	ObjectId('6a290785aa70267...	"D103"	"Dr. Priya"	"Endocrinology"	12	

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Patients

Cluster0 > hospitalManagementDB > patients

Documents 3

Aggregations

Schema

Indexes 1

Validation

View monitoring

Visualize Your Data

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 3 of 3

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patients

	_id ObjectId	patientId String	name String	age Int32	disease String	
1	ObjectId('6a290719aa70267...	"P101"	"Rahul Kumar"	35	"Fever"	
2	ObjectId('6a290726aa70267...	"P102"	"Meera Nair"	28	"Migraine"	
3	ObjectId('6a290734aa70267...	"P103"	"Arjun Das"	42	"Diabetes"	

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Assignment

Student Management Database

Database

studentManagementDB

Cluster0 > studentManagementDB

View monitoring

Visualize your data

Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
courses	-	20.48 kB	624.00 B	5	124.00 B	1	20.48 kB
students	-	36.86 kB	1.11 kB	10	111.00 B	1	36.86 kB

Collections Created

- 1. students
- 2. courses

Sample Documents

Students collection

Cluster0 > studentManagementDB > courses

View monitoring

Visualize Your Data

Documents5

Aggregations

Schema

Indexes1

Validation

Type a query: { field: 'value' } or Generate query

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT CODE

25

1 - 5 of 5

courses						
_id	ObjectId	courseId String	courseName String	duration String	instructor String	
1	ObjectId('6a2909f4aa70267...	"C101"	"Web Development"	"6 Months"	"Arun Kumar"	
2	ObjectId('6a290a01aa70267...	"C102"	"Python Programming"	"4 Months"	"Anjali Nair"	
3	ObjectId('6a290a0caa70267...	"C103"	"Data Science"	"8 Months"	"Rahul Menon"	
4	ObjectId('6a290a1aaa70267...	"C104"	"Java Programming"	"5 Months"	"Priya Das"	
5	ObjectId('6a290a26aa70267...	"C105"	"Mobile App Development"	"6 Months"	"Vishnu Raj"	

Courses Collection

Cluster0 > studentManagementDB > courses

Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	courseId String	courseName String	duration String	instructor String	
1	ObjectId('6a2909f4aa70267...')	"C101"	"Web Development"	"6 Months"	"Arun Kumar"	
2	ObjectId('6a290a01aa70267...')	"C102"	"Python Programming"	"4 Months"	"Anjali Nair"	
3	ObjectId('6a290a0caa70267...')	"C103"	"Data Science"	"8 Months"	"Rahul Menon"	
4	ObjectId('6a290a1aaa70267...')	"C104"	"Java Programming"	"5 Months"	"Priya Das"	
5	ObjectId('6a290a26aa70267...')	"C105"	"Mobile App Development"	"6 Months"	"Vishnu Raj"	

CRUD Operations Performed:

1.Create

- Created the database studentManagementDB.
- Created the students and courses collections.
- Inserted 10 student records and 5 course records.

2.Read

- Retrieved and viewed all student records from student collection where course is "BCA".
- Verified the stored data using MongoDB Atlas.

Cluster0 > studentManagementDB > students

Documents 10 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

{ "course": "BCA" }

Generate query Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 3 of 3

	_id ObjectId	studentId String	name String	email String	course String	
1	ObjectId('6a290905aa70267...')	"S101"	"Rahul"	"rahul@gmail.com"	"BCA"	
2	ObjectId('6a290952aa70267...')	"S105"	"Vishnu"	"vishnu@gmail.com"	"BCA"	
3	ObjectId('6a2909a7aa70267...')	"S108"	"Neha"	"neha@gmail.com"	"BCA"	

3.Update

- Updated the course of student S101 from BCA to BSc CS.
- Verified that the changes were saved successfully.

Cluster0 > studentManagementDB > students View monitoring Visualize Your Data

Documents 10 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options >

ADD DATA UPDATE DELETE EXPORT CODE 25 1 - 1 of 1 < > ≡ {} 🔍

	_id ObjectId	studentId String	name String	email String	course String
1	ObjectId('6a290905aa70267...	"S101"	"Rahul"	"rahul@gmail.com"	"BSc Cs"

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4.Delete

- Deleted the student record with Student name Rahul.
- Verified that the record was removed from the collection.

Cluster0 > studentManagementDB > students View monitoring Visualize Your Data

Documents 9 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options >

ADD DATA UPDATE DELETE EXPORT CODE 25 1 - 0 of 0 < > ≡ {} 🔍



No results

Try modifying your query to get results.

Conclusion

This assignment helped me understand how MongoDB Atlas is used to create databases and collections, store data, and perform CRUD operations. It also improved my practical knowledge of database management and data organization.